

Varshaa Sekar Jayanthi

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Software Engineer | 2+ years full-time in production C#/.NET and Python

EDUCATION

Northwestern University, McCormick School of Engineering

Chicago, IL

Master of Science in Information Technology | GPA: 3.75

Sep 2025 – Dec 2026

- Computer Science coursework: Machine Learning, Deep Learning, Scalable Software Architecture, Algorithms

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Information Systems | Minor in Business Administration

Aug 2019 – May 2023

- Computer Science coursework: 15-112 Fundamentals of Programming, 15-121 Data Structures (Java), 17-313 Software Engineering
- Awards: Grace Hopper Celebration Scholarship, Senior Leadership Recognition Award

EXPERIENCE

Tenacitics, Inc.

Houston, TX (Remote)

AI Engineer Intern | Internal SQL deployment platform

Jun 2026 – Present

- Cut deployment effort **68%** across **120** developers and DBAs, replacing a **3**-system ticket-and-email handoff with same-day deploys
- Eliminated partial writes across all **4** environments: transactional execution, enforced timeouts, API-level role separation
- Proved that a failed deployment leaves the database unchanged: **113** cases at **98%** coverage against live PostgreSQL 16, not mocks

Northwestern University

Chicago, IL

Graduate Researcher | Agentic Systems Evaluation | Contributor, paper in preparation

Jan 2026 – Jun 2026

- Lifted task accuracy **54%** to **78.5%**, the system's core result, by running the ablation isolating verifier-grounded memory across **400+** GPT-5.5 evaluations over **155** tasks
- Diagnosed environment setup, not model reasoning, as the dominant failure cause, then co-built the **10**-attempt recovery loop

Computer Enterprises Inc.

Pittsburgh, PA

Software Developer, embedded vendor-side owner | Client: Zelis Payments

Jun 2024 – Aug 2025

- Shipped **10** major features on a large-scale healthcare payments platform in C#/.NET and React, reducing bug backlog **35%**
- Improved database performance **40%** by rewriting slow queries, restructuring joins, and adding indexes
- Built an LLM pipeline that carried **10+** enterprise data migrations at **98%** accuracy and **20%** lower cycle time

UPMC Corporate Services

Pittsburgh, PA

Software Engineer

Jul 2023 – Jun 2024

- Developed **~20 features** on a Health Plan web application in C#/.NET against MS SQL Server, spanning new API endpoints, service integrations across UPMC systems, and stored procedures, schema changes and migrations
- Lowered production error rates **47%** with release gating; owned the lower-environment database and narrowed over-fetching APIs

Carnegie Mellon University

Pittsburgh, PA

Teaching Assistant | Application Design and Development

Jan 2023 – Aug 2023

- Taught weekly labs to **80+** students in a required core CMU course: MVC, relational database design, and full-stack Rails

PROJECTS & COMPETITIONS

BluePrint | Early-stage startup, AI-orchestrated SDLC governance platform; TypeScript, PostgreSQL, Express

Jul 2026 – Present

- Accelerated a knowledge-graph subgraph query from **31 s to 183 ms** and **472,421** intermediate rows to **123**, diagnosing fan-in convergence in a recursive CTE; output verified set-identical
- Closed **2** live cross-tenant data-exposure paths, one a state-changing write, by root-causing parameter loss in Express middleware; audited **22** routers for blast radius

IMC Prosperity 4 Global Algorithmic Trading Competition | Python

Apr 2026 – May 2026

- Team captain; reached the final round, ranking **1,275 of 18,803 teams** (top 7%), and doubled per-day profit on the highest-volume product by replacing a static fair value with a rolling-mean estimator

Prediction Market Order Book & Market-Making Simulator | Python; C++17 port in progress

May 2026 – Present

- Engineered an event-driven limit order book with price-time priority FIFO over live Kalshi prediction markets, using sorted price levels ($O(\log n)$ insert), per-level dequeues, and hash-map cancellation ($O(1)$)
- Reconciled a full P&L decomposition to **1e-12** against realized cash, separating spread capture, inventory drift, and fees

LEADERSHIP

Grew membership **80 to 200** as elected **Vice President, Activities Board**, Carnegie Mellon Student Government (2022 to 2023), owning budget, programming, and performance decisions.

Allocated **\$2M** annually across **250** student organizations against \$4.2M in requests as elected **Lead Financial Advisor, Joint Funding Committee**, Carnegie Mellon Student Government (2021 to 2023), leading 3 advisors.

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript/JavaScript, SQL, Java, Ruby, Swift

Frameworks & APIs: .NET, React, Node.js, Express, FastAPI, Rails, REST APIs, Agile/Scrum, CI/CD, Azure DevOps

Infrastructure & Data: AWS (Lambda, S3, API Gateway), PostgreSQL, MS SQL Server, MongoDB, Docker, Kubernetes, Linux, Git

ML & Data: LLM evaluation, regression, statistical analysis, NumPy, Pandas, scikit-learn